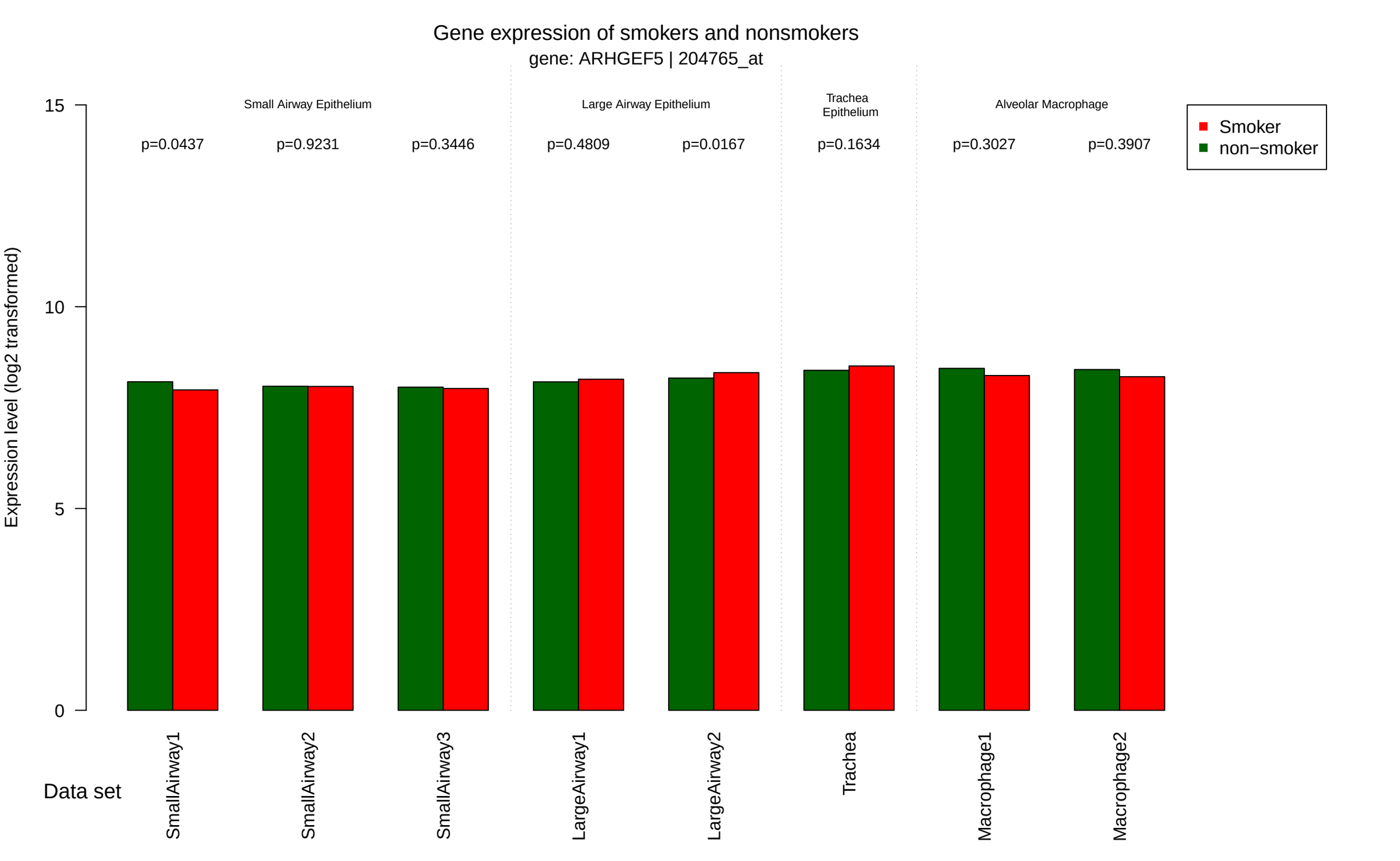
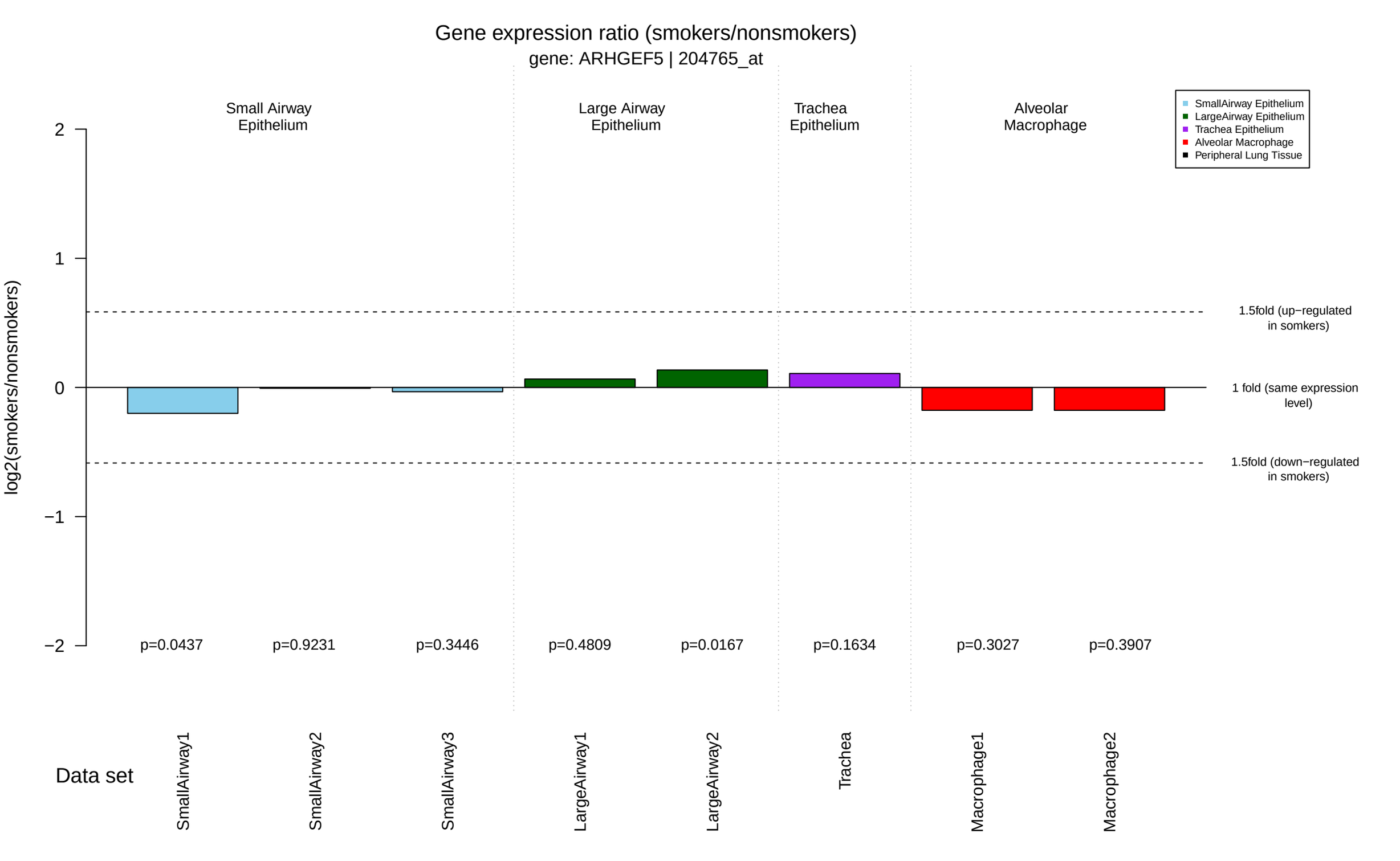
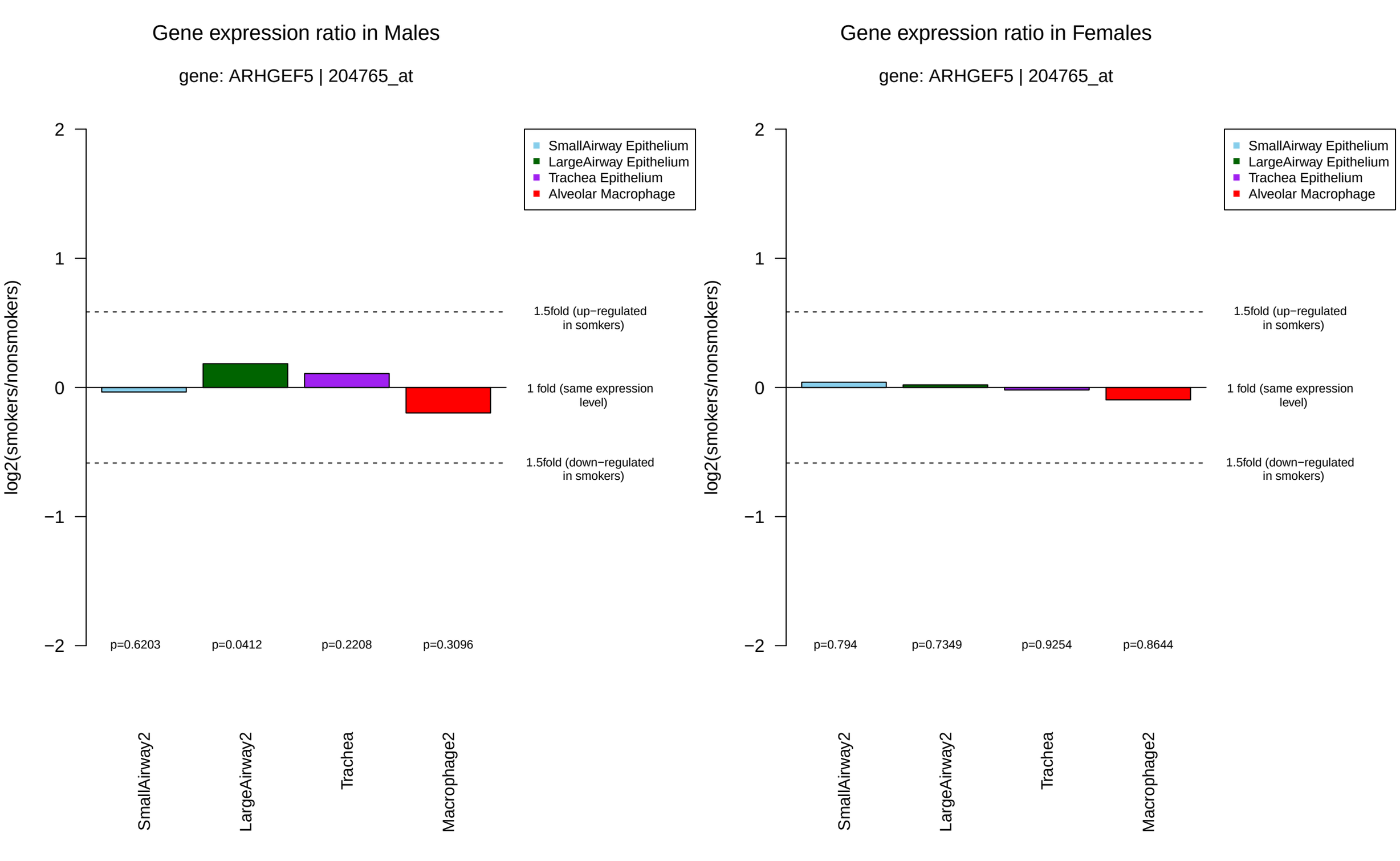
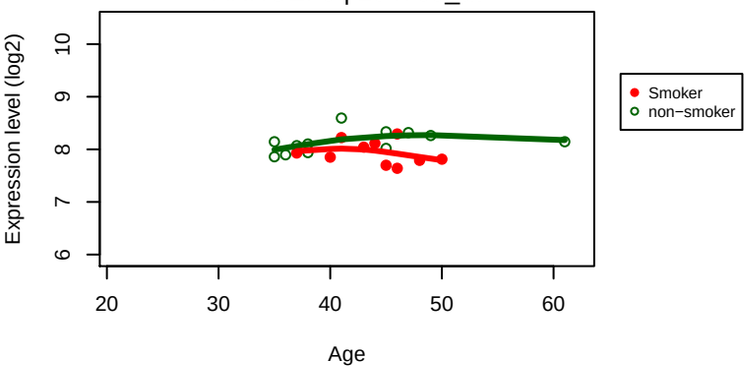
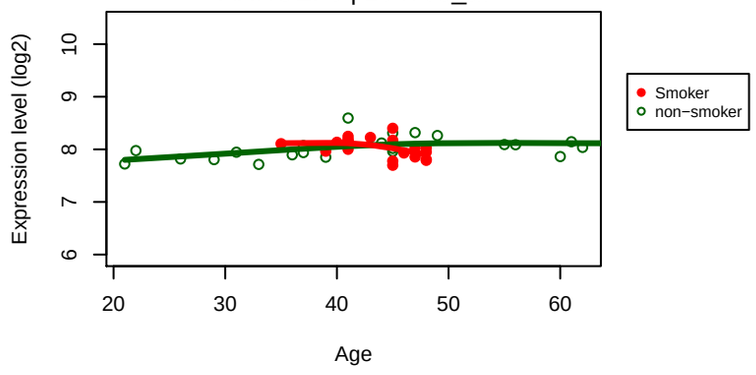
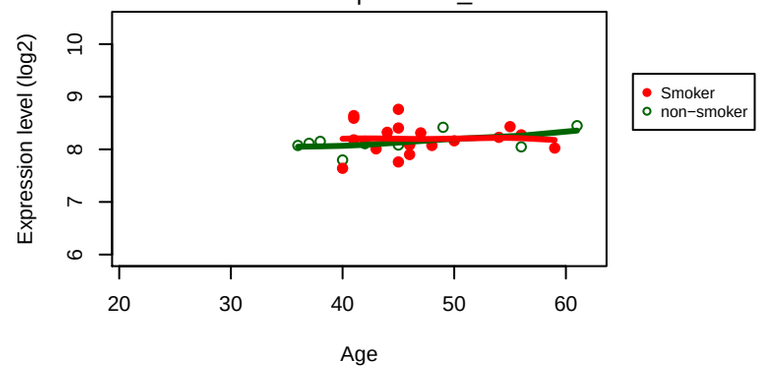

Gene expression of smokers and nonsmokers
gene: ARHGEF5 | 204765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0437
p=0.9231
p=0.3446
p=0.4809
p=0.0167
p=0.1634
p=0.3027
p=0.3907
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

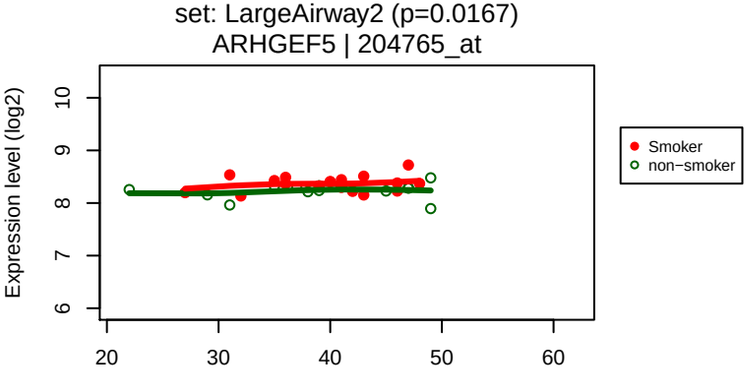

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF5 | 204765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0437
p=0.9231
p=0.3446
p=0.4809
p=0.0167
p=0.1634
p=0.3027
p=0.3907
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

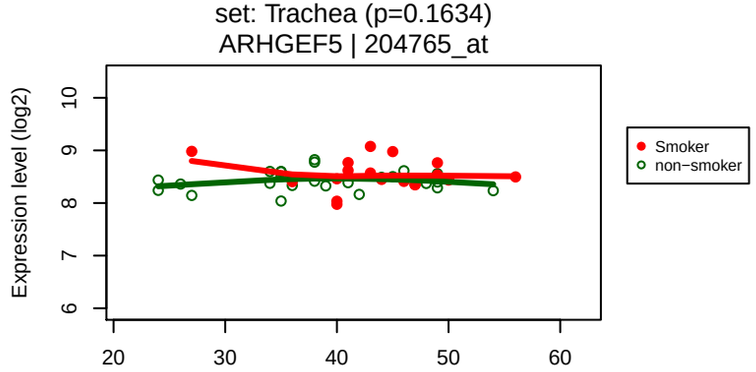

Gene expression ratio in Males
gene: ARHGEF5 | 204765_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6203
p=0.0412
p=0.2208
p=0.3096
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF5 | 204765_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.794
p=0.7349
p=0.9254
p=0.8644
SmallAirway2
LargeAirway2
Trachea
Macrophage2

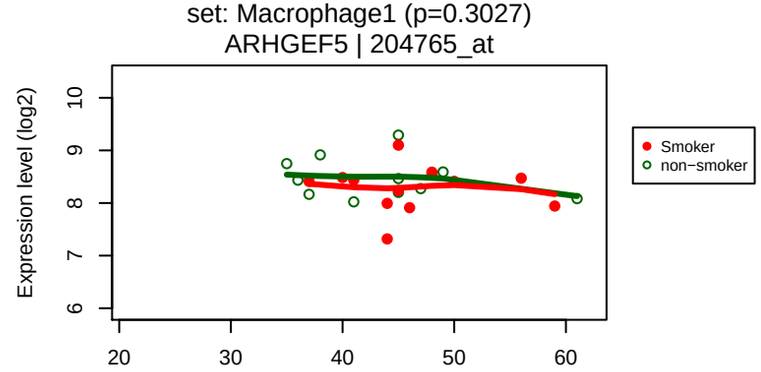

set: SmallAirway1 (p=0.0437)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker

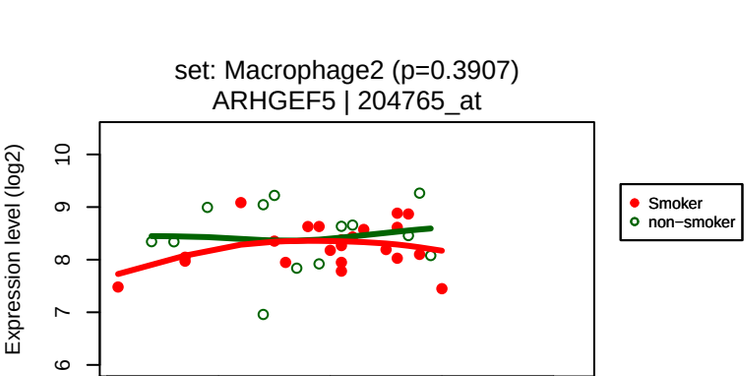

set: SmallAirway2 (p=0.9231)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker

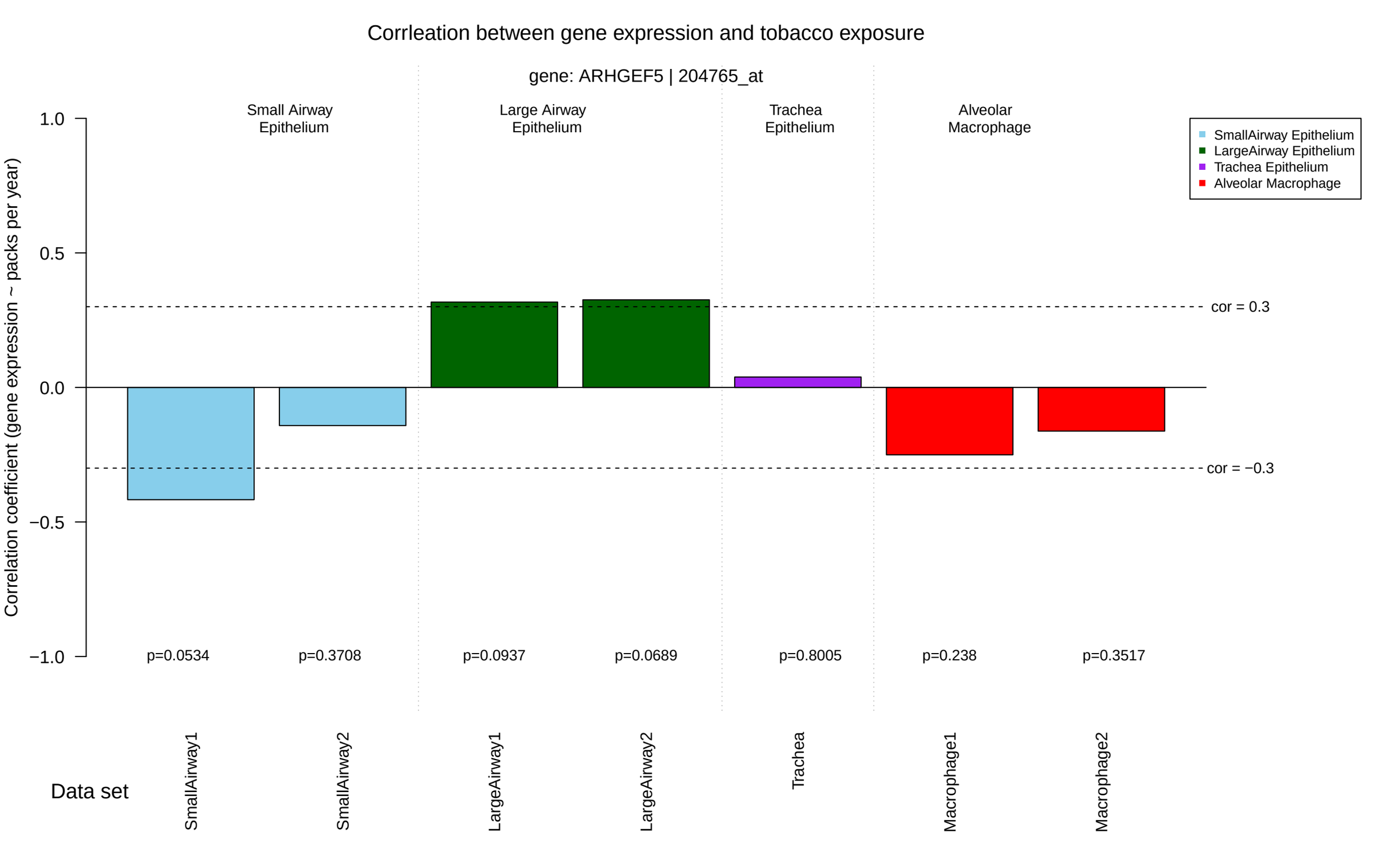

set: LargeAirway1 (p=0.4809)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0167)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1634)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3027)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3907)
ARHGEF5 | 204765_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF5 | 204765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0534
p=0.3708
p=0.0937
p=0.0689
p=0.8005
p=0.238
p=0.3517
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2